

U.S.A Airline Delay Analysis from 2023 - 2024

Flight Activity & Delay Trends

Delay Drivers & Airline Performance

Total Flight Arrivals

997K

Total Flight Delayed

170K

Total Flight Cancelled

10K

Total Flight Diverted

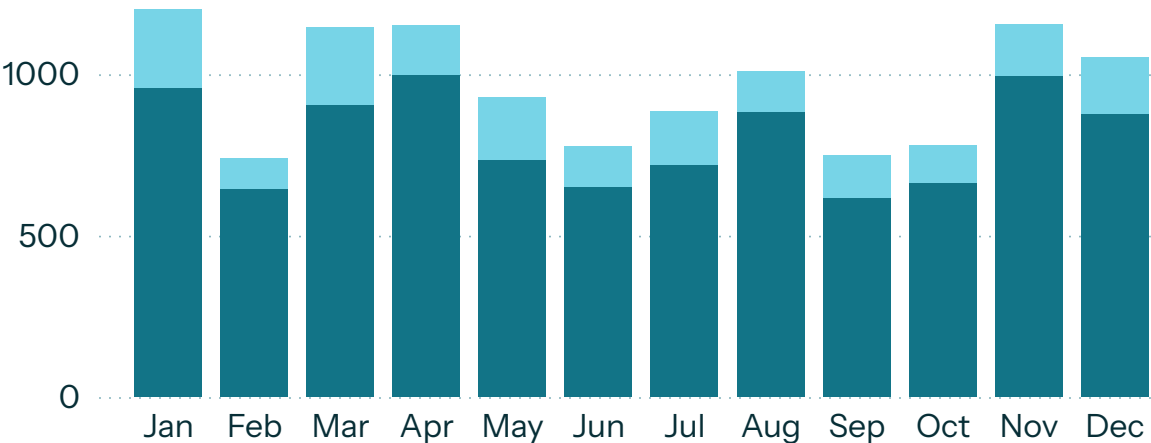
2K

On-Time Rate

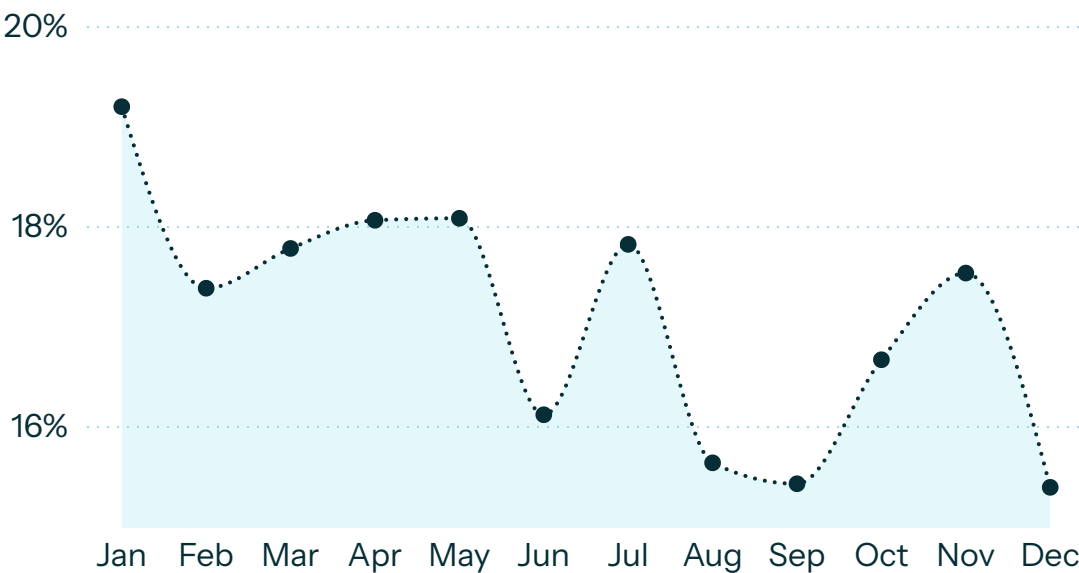
82.90%

Cancelled vs. Diverted Flights by Month

Cancelled Flights Diverted Flights

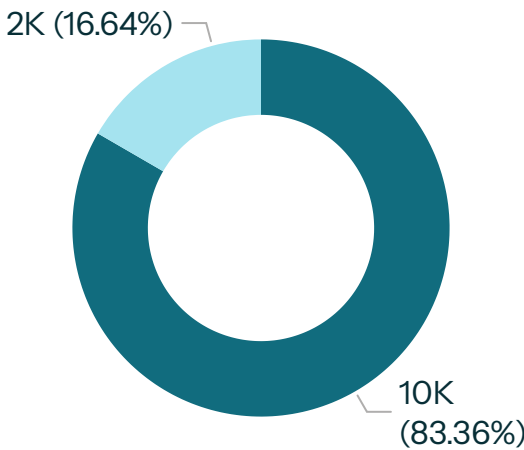


% of Delayed Flights per Month

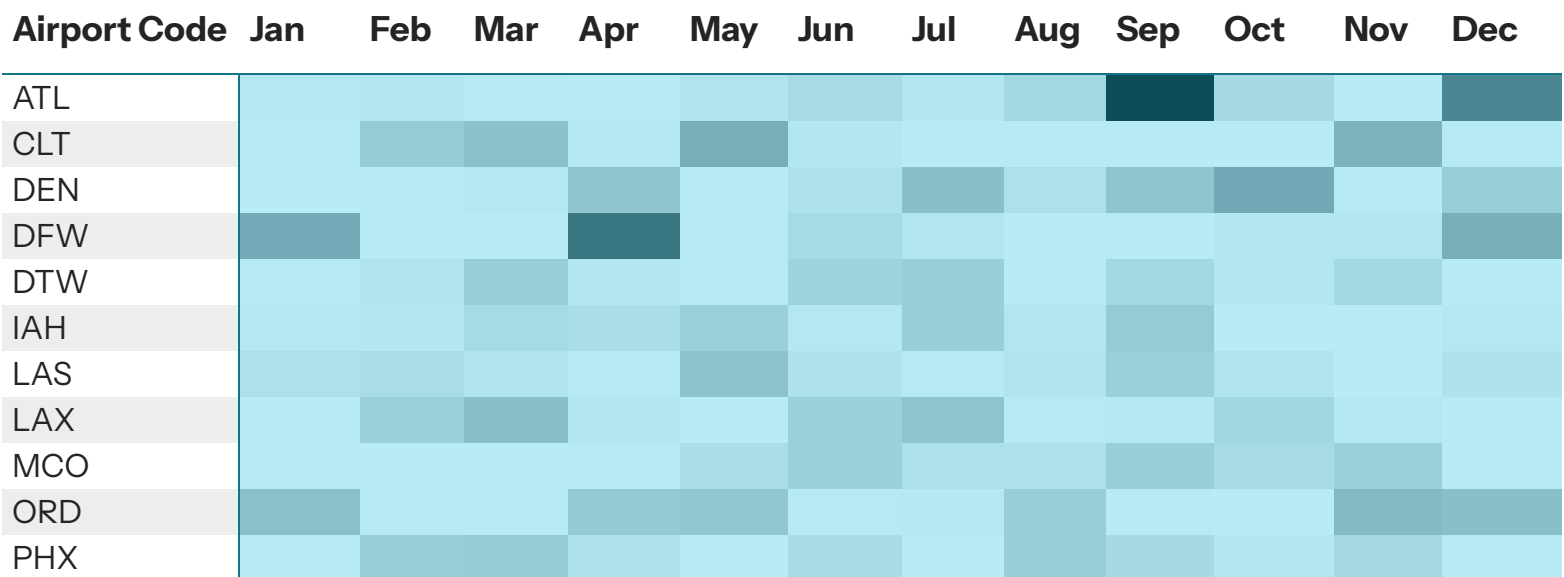


Cancelled vs Diverted Flights

Total Cancelled Flights Total Diverted Flights

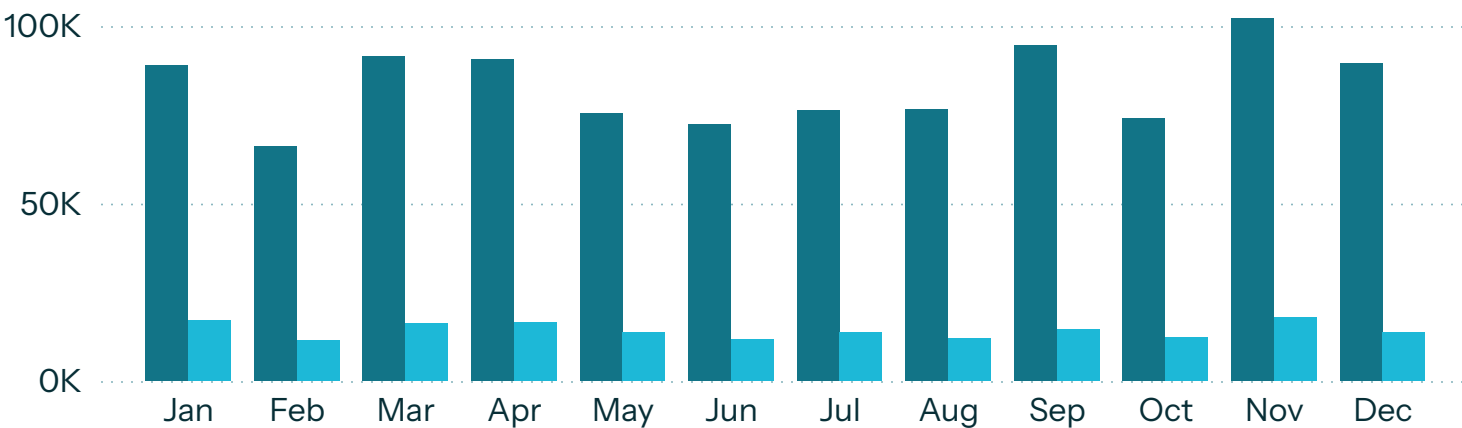


Busiest Airports by Flight Volume per Month



Total Flights vs. Delays by Month

Total Flights Arrival Total Delays



Pre-Analysis

Observations

Recommendations

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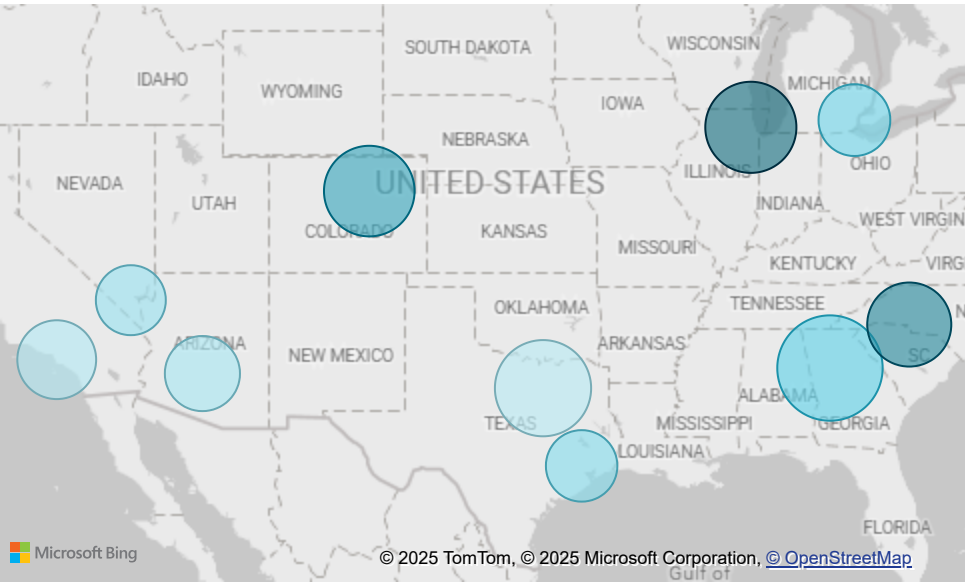
Total Flight Diverted

2K

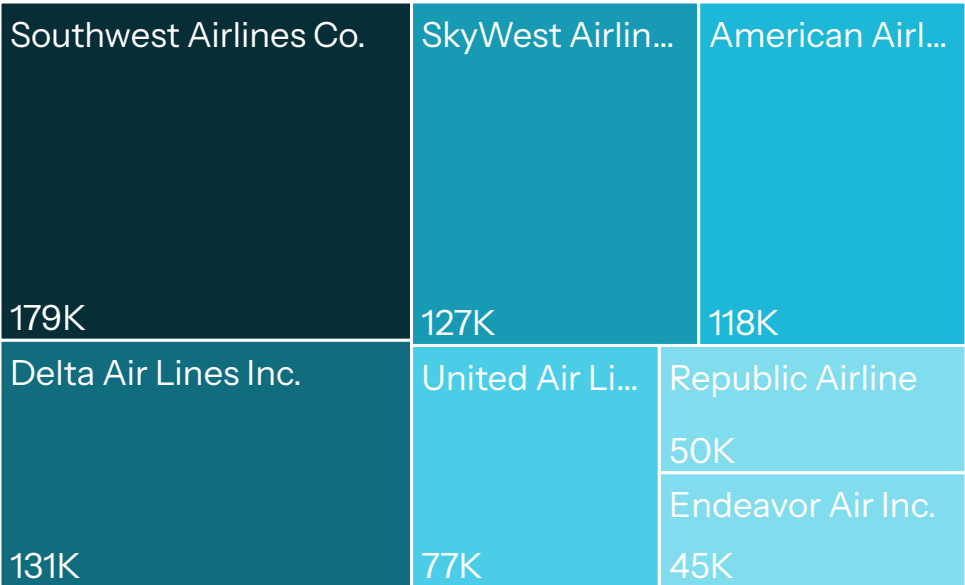
On-Time Rate

82.90%

Flight Volume by Location



Flight Volume by Carrier Name



Delay Impact Distribution by Airline

Carrier Name	Aircraft Delay (%)	Carrier Delay (%)	System Delay (%)	Weather Delay (%)	Security Delay (%)
Alaska Airlines Inc.	29.95%	28.33%	40.05%	1.30%	0.37%
Allegiant Air	36.79%	22.10%	31.36%	9.25%	0.48%
American Airlines Inc.	30.41%	30.41%	36.80%	2.09%	0.29%
Delta Air Lines Inc.	23.56%	30.62%	42.77%	2.88%	0.17%
Endeavor Air Inc.	36.96%	28.04%	31.79%	3.12%	0.10%
Envoy Air	36.01%	16.69%	41.58%	5.39%	0.32%
ExpressJet Airlines LLC	40.36%	19.62%	37.99%	2.03%	0.00%
Frontier Airlines Inc.	33.88%	28.55%	36.47%	1.10%	0.00%
Hawaiian Airlines Inc.	11.54%	87.22%	0.00%	1.23%	0.00%
JetBlue Airways	27.19%	36.91%	34.22%	1.55%	0.14%
Mesa Airlines Inc.	45.12%	21.77%	29.61%	3.24%	0.26%
PSA Airlines Inc.	47.65%	20.93%	29.02%	2.12%	0.28%
Republic Airline	31.06%	25.47%	40.11%	2.81%	0.55%
SkyWest Airlines Inc.	36.22%	37.38%	20.06%	6.06%	0.27%
Southwest Airlines Co.	40.41%	34.88%	23.44%	0.81%	0.46%
Spirit Air Lines	20.09%	24.47%	52.41%	2.09%	0.94%
United Air Lines Inc.	32.50%	25.98%	38.59%	2.91%	0.02%

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Pre-Analysis Board

Project Objectives

This project aims to uncover critical patterns and trends in airline delays across different airports, airlines, and time periods. By analysing flight volume, delay causes, and performance metrics, the goal is to provide a clearer understanding of what contributes most to delays and how these factors vary by location and season. Through this analysis, we intend to evaluate which airlines and airports maintain better on-time records, identify key operational bottlenecks such as late aircraft and weather disruptions, and highlight geographical or seasonal delay hotspots. Ultimately, the findings will support data-driven recommendations to improve flight scheduling, reduce delays, and enhance both passenger experience and operational efficiency in the aviation sector.

Key Metrics and Dimensions

Metrics (Measures):

Total Flights
Total Delayed Flights (>15 mins)
% Flights Delayed
Total Cancelled Flights
Total Diverted Flights
Delay Minutes by Cause (Carrier, Weather, System, Security, Late Aircraft)
Delay Rates by Airline and Airport

Dimensions (Filters/Slicers):

Year / Month
Airport Name / Code
Airline Name
Delay Cause
Peak vs Off-Peak Months

Pre-Analysis Board

Project Objectives

1. What are the busiest months and airports for flights?
2. Which months or seasons experience the highest delay rates?
3. Which airlines have the highest or lowest delay rates?
4. What are the top reasons for delays across the board?
5. Are certain airports more prone to weather-related delays?
6. Is there a pattern in delays between coastal and inland airports?
7. What is the percentage of flights delayed over 15 minutes?
8. Which carriers or airports perform best in on-time delivery?
9. Do certain airports have consistently high cancellation or diversion rates?

Key Metrics and Dimensions

1. Summer and holiday travel months (e.g., July, December) show the highest flight volumes and delay rates.
2. A significant portion of delays is caused by late-arriving aircraft, indicating a cascading effect in operations.
3. Carrier-related delays are highest among low-cost carriers, suggesting internal resource constraints.
4. Major hubs like Airport X and Airport Y experience high volumes but manage low delay percentages due to strong infrastructure.
5. Weather delays are regionally concentrated, especially in northern and coastal airports during winter and storm seasons.
6. Some regional airports suffer high cancellation rates despite low volume, possibly due to dependency on connecting flights.
7. Airlines that maintain fewer diverted or canceled flights show stronger scheduling and contingency planning.
8. Off-peak months (e.g., February, September) see fewer delays and better operational efficiency.
9. Certain carriers consistently outperform others in on-time performance, offering potential best-practice models.

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Observations Board

1. Aircraft-related delays are disproportionately high for PSA Airlines (47.65%) and Mesa Airlines (45.12%).

This suggests mechanical or operational inefficiencies with aircraft readiness.

2. Hawaiian Airlines shows extremely high carrier-related delays (87.22%) despite low aircraft and system delay contributions.

Indicates possible internal crew, maintenance, or scheduling issues.

3. Delta and United Airlines experience a large portion of delays due to system issues (Delta: 42.77%, United: 38.59%).

Points to infrastructure or logistical challenges beyond direct airline control.

4. Allegiant Air has a notably high weather delay rate (9.25%)—the highest among all carriers.

Could imply route concentration in weather-prone regions or poor contingency routing.

5. December has the highest number of cancellations (876) and a high volume of diverted flights (176).

Likely due to winter storms and holiday travel surges.

Observations Board

6. July is one of the busiest months by volume (76,071 flights) yet maintains a mid-range delay percentage (17.82%).

Suggests well-managed traffic despite seasonal spikes.

7. November has the highest volume of delayed flights over 15 minutes (17,920) and the highest cancellation count (994).

Holiday travel peaks and weather may be overwhelming airport capacity.

8. The overall delay rate remains consistently between 15–19% throughout the year, with slight dips in Sep (15.42%) and Dec (15.39%).

This implies systemic scheduling challenges that persist year-round.

9. ATL (Atlanta) is the busiest airport with over 55,000 flights/month, yet maintains a relatively stable delay profile.

Shows high operational efficiency despite large traffic volumes.

10. SkyWest Airlines exhibits a balanced distribution of delay causes but a moderately high weather delay rate (6.06%).

Indicates exposure to seasonal or regional weather fluctuations.

Pre-Analysis

Observations

Recommendations

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Recommendations Board

- 1. I recommend that PSA and Mesa Airlines prioritize major investments in aircraft maintenance and reduce turnaround time delays.
We need to minimize technical issues that consistently cause nearly half of their delays.
- 2. Hawaiian Airlines needs to conduct a thorough operational audit to identify root causes of its high carrier-related delays.
I recommend improving crew scheduling, internal logistics, and maintenance planning.
- 3. Delta and United should work closely with airport authorities to resolve system-related delays.
We need to streamline processes like baggage handling, gate assignments, and taxiway logistics.
- 4. I recommend that Allegiant Air re-evaluate its route network and implement weather contingency plans.
We need to proactively manage weather-related risks through smarter routing and better preparedness.
- 5. For December, I recommend airlines activate winter contingency strategies.
We need to deploy extra de-icing teams, allocate buffer time in flight schedules, and have standby aircraft ready.

Recommendations Board

- 6. We should study July's performance as a benchmark for managing high volumes with minimal delays.
I recommend documenting successful practices and using them to guide operations in busier months.
- 7. Airports and airlines need to boost staffing and implement digital crowd management tools in November.
I recommend increasing operational readiness during the Thanksgiving and holiday rush.
- 8. I recommend using predictive delay modelling tools to optimize year-round scheduling.
We need AI-powered systems to anticipate delays and adjust plans proactively.
- 9. We should treat ATL's scheduling model as a case study for operational efficiency at high-traffic airports.
I recommend other major airports adopt and test similar logistics systems.
- 10. SkyWest should enhance its weather monitoring systems and readiness strategies.
We need to reduce weather-related delays by being more regionally responsive and better prepared.